1. System-Prompt Decoupling

Coupled — × Action collapse

Full 10-Action Agent Prompt + Click-Only Grounding Data

click ✓ | type × | scroll × | key ×

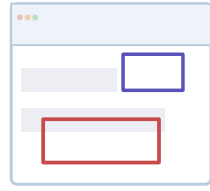
Decoupled (Ours) — ✓ All actions active

Click Prompt enum: ["click"] → Click Data

Full Prompt 10 GUI Actions → Trajectories

click ✓ | type ✓ | scroll ✓ | key ✓

2. Same-Image Packing & Dedup



Q1: "New File" → A1 [351, 378]

Q2: "Settings" → A2 [163, 781]

Q3: "Where is Settings?" → A2

① Unpacked — Img | Q1 | A1 | Img | Q2 | A2

3× image tokens (slow)

② Raw Pack — Img | Q1 | A1 | Q2 | A2 | Q3 | A3=A2

× A3 copies A2 in context

③ Dedup (Ours) — Img | Q1 | A1 | Q2 | A2

✓ 1× image, zero copying

3. Reasoning vs. Grounding Loss

Loss tokens per sample (to scale):

Grounding: Prompt+Img | 33 tok (click)

Tool Call: Context | <think> 397 (69%) | Call 178

× 69% of tool-call gradient is <think> prose

Recipe A (Adopted): Selective <think> Masking

<think> | masked (w=0) | </think> | Call

✓ Keeps tags & focuses loss on tool call

Recipe B: Early GUI Trajectory Mixing

Mixes SFT GUI trajectories into Stage 1
(similar final score; Recipe A is faster & lighter)